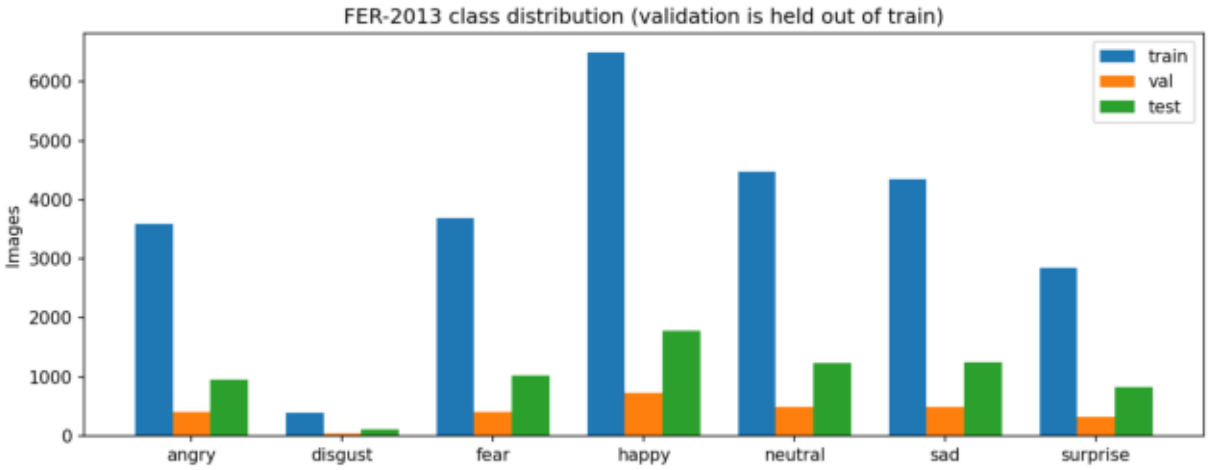
Facial emotion recognition - methods

Data: msambare/fer2013 on Kaggle. Seven emotions; grayscale 48 x 48 pixels. No supplementary training
dataset is used. ResNet-18 starts from ImageNet-1K weights.

Split protocol: validation is stratified from the original training folder; the supplied test folder is reserved for final evaluation. The saved split manifests record every file and label. Model selection uses validation accuracy, with macro F1 as a tie-breaker.

Split sizes: train=25,838, val=2,871, test=7,178.
Class order: angry, disgust, fear, happy, neutral, sad, surprise.



Preprocessing: convert to grayscale, resize to 48 x 48, scale pixels to [0,1], then normalize with mean=0.5 and std=0.5. Training adds horizontal flips and mild affine rotation, shift and scale. Validation, test and realtime inference use no augmentation. Square-root inverse class-frequency loss
weights are derived only from training labels.

M1 (from scratch): four Conv3x3-BatchNorm-ReLU-MaxPool2x2-Dropout blocks, with 32/64/128/256 channels
and dropout 0.10/0.15/0.20/0.25. Flatten 3x3x256=2304, Linear256-BatchNorm-ReLU-Dropout0.50, then
Linear7.

M2 (transfer): ImageNet ResNet-18; sum the first convolution's RGB weights to accept one channel, set
its stride to 1 and remove the initial max pool to preserve small-image detail. Replace the head with
Dropout0.30-Linear7 and fine-tune every layer. The input normalization is deliberately adapted for FER, rather than the stock ImageNet transform.

Both models return logits for cross entropy; softmax is applied at inference.

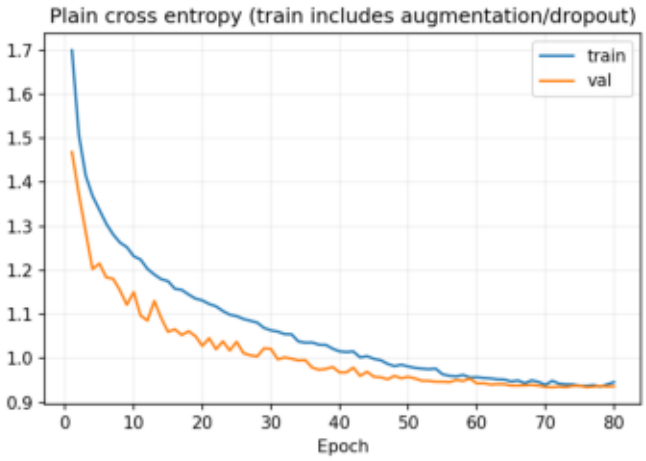
Training and quantitative comparison

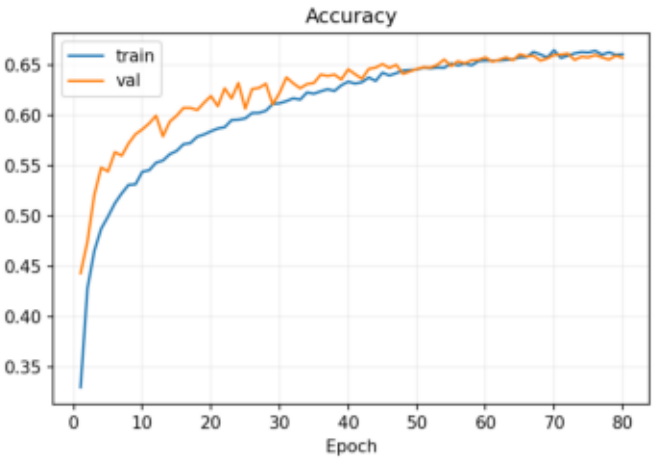
CNN: AdamW lr=0.001, backbone lr=n/a, weight decay=0.0001, smoothing=0.05, epochs <= 80, patience=18,
seed=42. Cosine LR schedule. Best epoch=72. Training device=cuda.

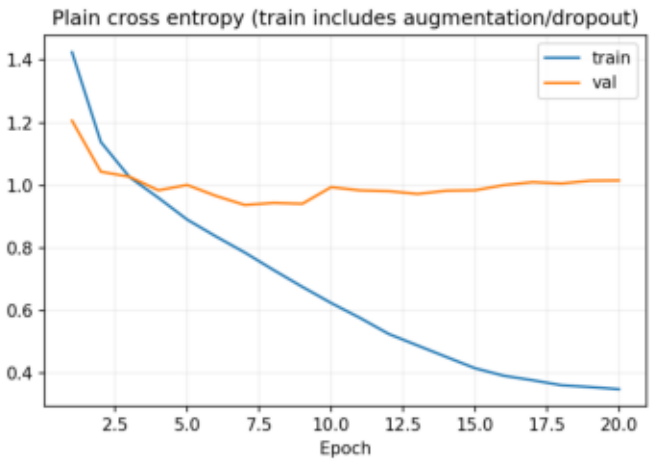
TRANSFER: AdamW lr=0.001, backbone lr=0.0001, weight decay=0.0001, smoothing=0.05, epochs <= 20,
patience=12, seed=42. Cosine LR schedule. Best epoch=18. Training device=cuda.

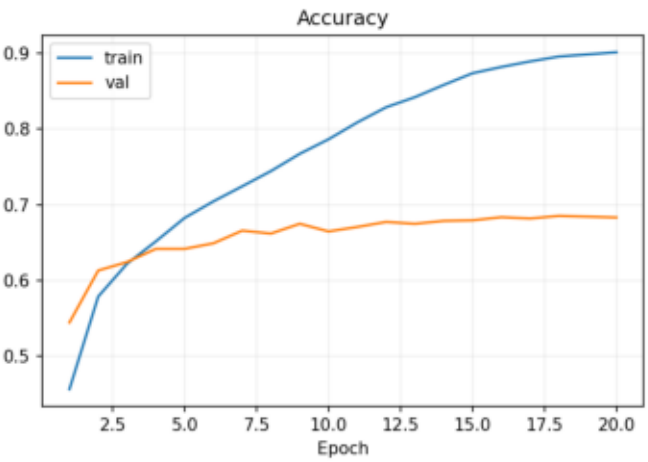
Model	Accuracy	Macro F1	Params	CPU FPS	Target met
cnn	65.99%	0.6337	980,711	449.6	yes
transfer	67.69%	0.6577	11,173,831	36.2	yes

Targets: M1 >= 60%, M2 >= 65%. CPU FPS is batch-one model forward only (float32, 1x48x48); it excludes preprocessing, face detection and camera I/O. Realtime throughput must be measured separately on the demonstration laptop. Benchmark CPU: x86_64, threads=4.

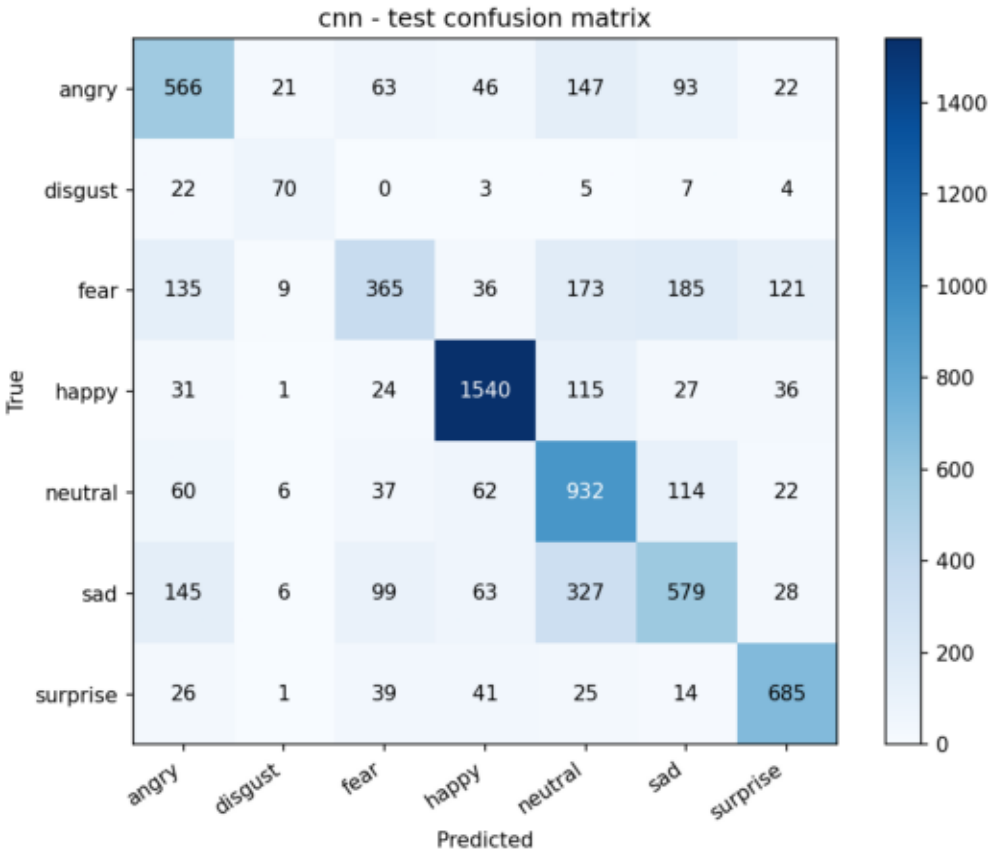


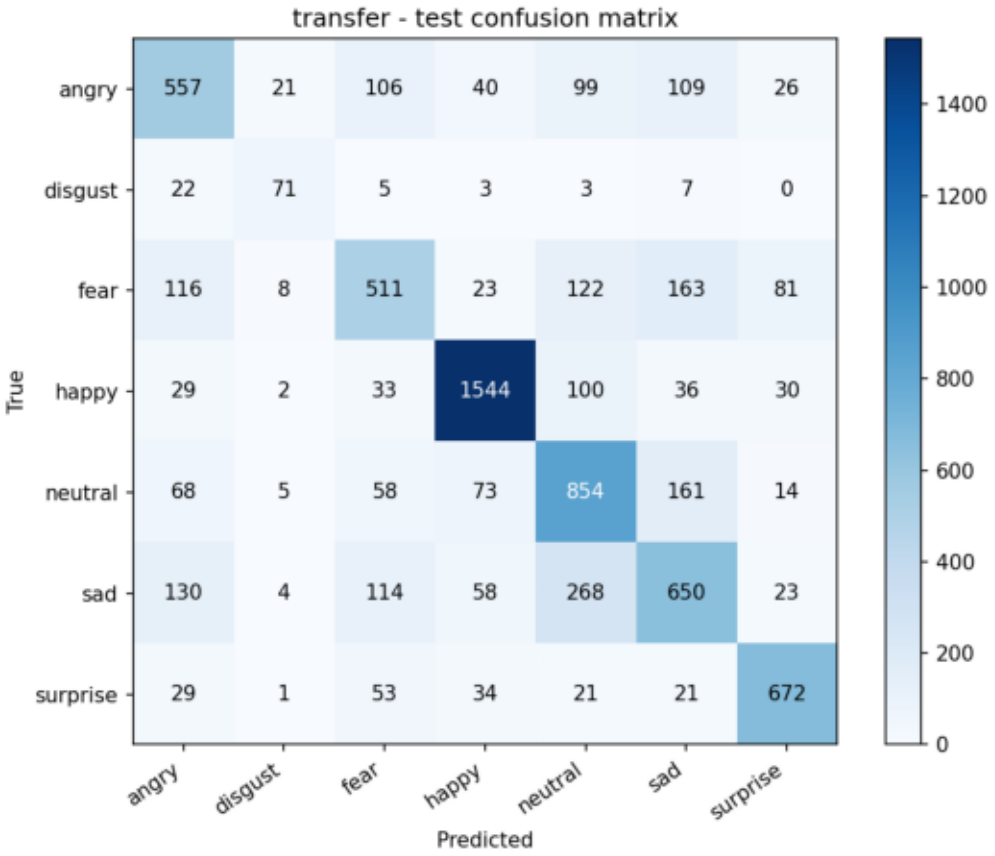






Confusion matrices on held-out test data





CNN - misclassified faces and analysis



1. angry -> sad. Assigned sad 78.5%, versus angry 3.0% (gap 75.5%). Inspect the shared facial cues and whether this label is visually ambiguous.

2. disgust -> surprise. Assigned surprise 66.6%, versus disgust 2.2% (gap 64.4%). Inspect the shared facial cues and whether this label is visually ambiguous.

3. fear -> angry. Assigned angry 42.2%, versus fear 32.8% (gap 9.4%). Inspect the shared facial cues and whether this label is visually ambiguous.

4. happy -> neutral. Assigned neutral 77.9%, versus happy 5.1% (gap 72.8%). Inspect the shared facial cues and whether this label is visually ambiguous.

5. neutral -> angry. Assigned angry 59.7%, versus neutral 15.4% (gap 44.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

6. sad -> fear. Assigned fear 32.4%, versus sad 16.0% (gap 16.4%). Inspect the shared facial cues and whether this label is visually ambiguous.

7. surprise -> happy. Assigned happy 55.3%, versus surprise 30.1% (gap 25.1%). Inspect the shared facial cues and whether this label is visually ambiguous.

8. angry -> sad. Assigned sad 36.9%, versus angry 25.2% (gap 11.7%). Inspect the shared facial cues and whether this label is visually ambiguous.

9. angry -> neutral. Assigned neutral 55.3%, versus angry 12.6% (gap 42.8%). Inspect the shared facial cues and whether this label is visually ambiguous.

10. angry -> disgust. Assigned disgust 39.1%, versus angry 20.9% (gap 18.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

Observations above describe score errors and suggest visual checks. Add your own image-specific observations before submission; score patterns alone cannot establish why a face was misclassified.

TRANSFER - misclassified faces and analysis



1. angry -> sad. Assigned sad 46.4%, versus angry 17.4% (gap 29.0%). Inspect the shared facial cues and whether this label is visually ambiguous.

2. disgust -> fear. Assigned fear 82.4%, versus disgust 1.0% (gap 81.4%). Confident error; inspect label ambiguity, face crop and domain mismatch.

3. fear -> angry. Assigned angry 31.2%, versus fear 25.0% (gap 6.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

4. happy -> fear. Assigned fear 30.5%, versus happy 18.8% (gap 11.7%). Inspect the shared facial cues and whether this label is visually ambiguous.

5. neutral -> fear. Assigned fear 49.7%, versus neutral 44.0% (gap 5.7%). Inspect the shared facial cues and whether this label is visually ambiguous.

6. sad -> surprise. Assigned surprise 85.7%, versus sad 0.1% (gap 85.6%). Confident error; inspect label ambiguity, face crop and domain mismatch.

7. surprise -> fear. Assigned fear 52.7%, versus surprise 20.5% (gap 32.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

8. angry -> neutral. Assigned neutral 32.8%, versus angry 16.7% (gap 16.1%). Inspect the shared facial cues and whether this label is visually ambiguous.

9. angry -> surprise. Assigned surprise 46.6%, versus angry 20.0% (gap 26.6%). Inspect the shared facial cues and whether this label is visually ambiguous.

10. angry -> neutral. Assigned neutral 47.8%, versus angry 19.2% (gap 28.6%). Inspect the shared facial cues and whether this label is visually ambiguous.

Observations above describe score errors and suggest visual checks. Add your own image-specific observations before submission; score patterns alone cannot establish why a face was misclassified.

Realtime system and demonstration

Pipeline: OpenCV VideoCapture -> Haar frontal-face detection -> crop each face with a small margin ->
shared grayscale preprocessing -> batched PyTorch inference -> softmax -> bounding boxes, labels,
confidence and FPS.

The command line selects cnn/transfer and accepts a phone URL, webcam, image or file video. Connection
failures trigger bounded retries and a clear error. Low-confidence predictions are labelled uncertain.
Press s to save the phone-camera demonstration.

The on-screen FPS includes capture and processing, but excludes display and output encoding. It depends on image resolution, number of faces, CPU threads and Wi-Fi. The 10 FPS CPU requirement must
be checked on the student's own laptop.

DEMONSTRATION PENDING: no real phone-camera screenshot was supplied. Run the app, press s, then
regenerate this report with --screenshot. This report is a draft until the demonstration and visual error analysis are completed.

Limitations: FER labels can be ambiguous; a visible expression does not reliably reveal a person's internal emotional state. Haar detection is strongest on frontal faces. Lighting, pose, occlusion and the FER-to-phone domain shift can reduce accuracy. Softmax scores are model confidence values, not calibrated probabilities.

Sources: kaggle.com/datasets/msambare/fer2013; torchvision ImageNet ResNet-18 (docs.pytorch.org/vision/stable/models/generated/torchvision.models.resnet18.html). Checkpoints, split manifests, class counts, run configuration, CSV metrics and error examples accompany the experiment for reproducibility.